

Canvas Engineering: Declared Causal Macrostructure for Reverse-Diffusion Latent Dynamics

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Abstract

Prompt engineering structures what a language model *sees*; canvas engineering structures what a diffusion model *thinks in*. The practitioner declares the macrostructure of a diffusion transformer’s latent space—which regions carry which modalities, their geometry, temporal update frequencies, loss participation, and, critically, a directed graph of *permitted* block-to-block attention operations—and a compiler lowers that declaration into attention masks, loss weights, and frame mappings. Because reverse diffusion is an iterated denoising map over the whole canvas, a hard connectivity constraint on attention induces an explicit, human-legible causal interaction graph *inside* the generative dynamics, while gradient descent remains free to shape all fine structure within and along the declared edges. This is a neurosymbolic architecture in a precise sense: transparent symbolic macrostructure—a type system, with offsets, signatures, calling conventions, and an ABI—hosted directly within learnable neural mechanics, with no discrete/continuous interface to cross. We give the underlying mathematics—region index sets, the mask construction, the weighted denoising objective, and the reachability semantics that make declared conditional independence exact—and describe the abstraction stack (regions $\rightarrow$ topology $\rightarrow$ typed programs $\rightarrow$ compiled deployment), including the compilation of nested entity schemas into hierarchically coarse-grained canvases. Worked designs span robot manipulation, a compiler-allocated hospital ICU ward with 199 regions, air-traffic control, and a 23-region cortical model that predicts real brain dynamics at $R^2 = 0.825$. Early experiments—26 studies, 236 training runs on CogVideoX-2B—include a 1.73 $\times$ parameter-efficiency result for looped attention ($p < 0.001$) and a frozen 350K-parameter configuration that beats 11.7M unfrozen parameters on action prediction. We are explicit that these are small-scale results and read them as calibration rather than narrative. We argue that *declared* latent structure is the production-ready on-ramp to intuition-guided symbolic world modeling: the symbolic model is not synthesized by the network, it is authored by the engineer, and the network learns everything else.

1 Position: two routes to symbolic world models

Chollet’s convergence thesis [1] holds that much of AI trends toward *intuition-guided symbolic world modeling*: deep-learning-guided program synthesis, because a symbolic model constructs a compact, reusable, highly generalizable representation of a problem from minimal data [2]. This inherits a long argument that con-

nectionism alone under-generalizes and that structure must re-enter the loop [3, 4]. The thesis names the destination but underdetermines the route.

The dominant proposed route is **synthesis**: a neural policy emits symbolic programs and is rewarded on their execution. It gains compactness and generalization, but pays for a brittle discrete search and—more corrosively—a hard boundary where continuous representations must be quantized into symbols and back. Canvas engineering takes the dual route, **declaration**: the human authors the symbolic model up front, not as a program the network emits but as the *interaction topology of latent space itself*, and the neural substrate fills it in. Both routes buy the same asset, a macrostructure acquired without data. Declaration ships today, because it compiles to nothing more exotic than attention masks and loss weights over an off-the-shelf pretrained diffusion transformer [5, 6].

The framing is economic. Neurosymbolic methods stay research curiosities as long as the marginal dollar buys more from data than from structure—a corollary of the bitter lesson [7]. Declared structure changes that calculus precisely where data is expensive: robotics, control, multi-agent coordination, world modeling [8–10]. A schema is free; a demonstration is not. This paper answers a prior question: *how does one make structure declarable at all* inside a modern generative architecture, and what does it buy empirically?

2 Mechanism

2.1 The canvas

A **canvas** is a 3D spatiotemporal grid (T, H, W) of d -dimensional latent positions, flattened into the token sequence of a diffusion transformer (DiT). A CanvasLayout partitions it into named regions (Fig. 1):

```
layout = CanvasLayout(  
    T=5, H=8, W=8, d_model=256,  
    regions={  
        "visual": (0,5, 0,6, 0,6), # 180 pos - video  
        "action": (0,5, 6,7, 0,1), # 5 pos - actions  
        "reward": (2,3, 7,8, 0,1), # 1 pos - scalar r  
    },  
    t_current=2, # t >= 2 is future (diffusion output)  
)
```

Each region carries a RegionSpec: its bounds; a temporal period mapping canvas frames to real frames, so a thought region at period=4 and a perception region at period=1 coexist on one canvas; is_output

CanvasLayout(T=5, H=8, W=8), $t_{\text{current}} = 2$

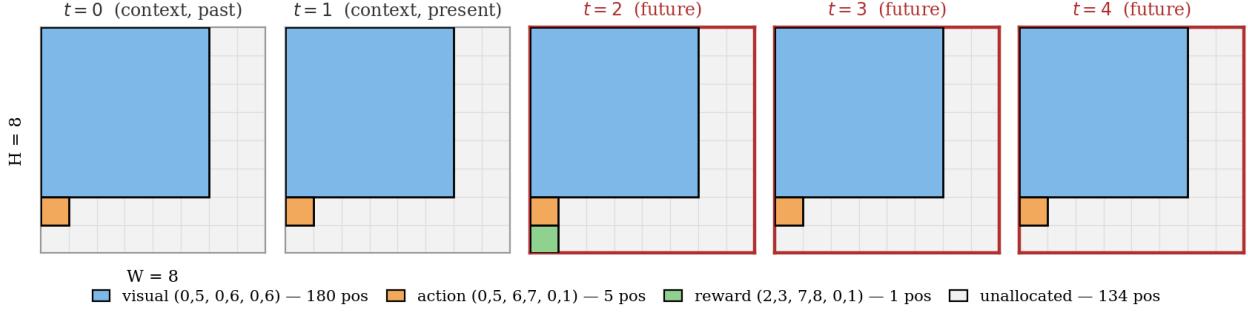


Figure 1: The exact layout declared by the code in §2.1, rendered as its five (H, W) time slices. `visual` occupies the 6×6 block at every frame (180 positions), `action` one position per frame (5), and `reward` a single position at $t = 2$. Frames $t \geq t_{\text{current}} = 2$ are future: denoised by reverse diffusion; earlier frames are clamped context. Unallocated positions (gray) carry no loss and no meaning.

and `loss_weight`, which make loss participation type-directed codegen—`loss_weight_mask()` compiles the declaration into a per-position weight tensor; an optional `semantic_type` with a frozen embedding; and a default `attn` function family. Heterogeneous modalities enter and exit through per-region encoders and decoders; the pretrained video backbone attends over everything. The canvas is thus the omnimodal I/O layer that lets a video model also *do things*—predict actions, read proprioception, estimate reward—rather than only generate pixels.

2.2 Topology: the causal graph is the attention mask

A `CanvasTopology` is a directed multigraph of discrete cross-attention operations. Each `Connection(src, dst)` licenses `src` tokens to query `dst` keys and values; the *absence* of an edge is a hard prohibition, not a soft prior:

```
CanvasTopology(connections=[
    # action queries visual: obs → act
    # information flow (diffusion policy)
    Connection(src="action", dst="visual"),
    # action at t queries visual at t-1
    Connection(src="action", dst="visual",
               t_src=0, t_dst=-1),
    # cross-agent coordination
    Connection(src="r1_cam", dst="r2_cam",
               weight=0.5),
])
```

Temporal offsets ($t_{\text{src}}, t_{\text{dst}}$) restrict which frame pairs participate—same-frame-only, previous-frame causal, sliding window—and a `TemporalFill` mode (HOLD/DROP/INTERPOLATE) resolves cross-frequency queries in real-time space when a `period=1` region queries a `period=4` one. Convenience constructors (`dense`, `isolated`, `hub_spoke`, `causal_chain`, `causal_temporal`) name the standard patterns (Fig. 2); a standard transformer [11] is the degenerate `dense` case.

This yields the paper’s central claim. Reverse diffusion applies the denoiser to the full canvas at every step [12, 13], so *which regions can influence which* under iteration is exactly the topology’s transitive closure. Declaring the topology therefore declares a causal inter-

action graph over the generative dynamics—attention-based rather than convolutional, so the graph is over *semantic regions* rather than spatial neighborhoods, and non-Euclidean by construction. Diffusion policy [14], observation→action, is the two-node base case. Multi-agent perceptual diffusion—each agent self-attending over its own observations and actions, coordinating only through declared cross-edges—is the same primitive composed.

What is fixed and what is learned splits cleanly. The engineer fixes *whether* an edge exists, its direction, temporal extent, and function family; gradients—from data or intrinsic losses—determine *what flows* along it and everything within regions. Macro is symbolic, micro is neural, and there is no interface to cross, because the symbolic layer *is* the mask.

2.3 Core mathematics

Four small pieces of arithmetic carry the whole construction.

Regions are index sets. A canvas has $N = T \cdot H \cdot W$ positions. A region r with bounds $(t_0, t_1, h_0, h_1, w_0, w_1)$ owns

$$I_r = \{ tHW + hW + w : t_0 \leq t < t_1, h_0 \leq h < h_1, w_0 \leq w < w_1 \},$$

with regions pairwise disjoint. This is struct-offset arithmetic, nothing more.

The topology compiles to a mask. Edges $E = \{(r_k \rightarrow s_k, w_k, \tau_k)\}$ lower to $M \in \mathbb{R}_{\geq 0}^{N \times N}$:

$$M_{ij} = \max_k w_k \mathbf{1}[i \in I_{r_k}] \mathbf{1}[j \in I_{s_k}] A_{\tau_k}(t(i), t(j)),$$

where A_τ is the temporal alignment predicate: for offsets $(t_{\text{src}}, t_{\text{dst}})$, positions align iff $\exists \text{ref} : t(i) = \text{ref} + t_{\text{src}}, t(j) = \text{ref} + t_{\text{dst}}$ (an unset offset is unconstrained). When periods differ, alignment is evaluated in real time $\rho_r(t) = p_r t$, and gaps resolve by fill mode: `HOLD` snaps

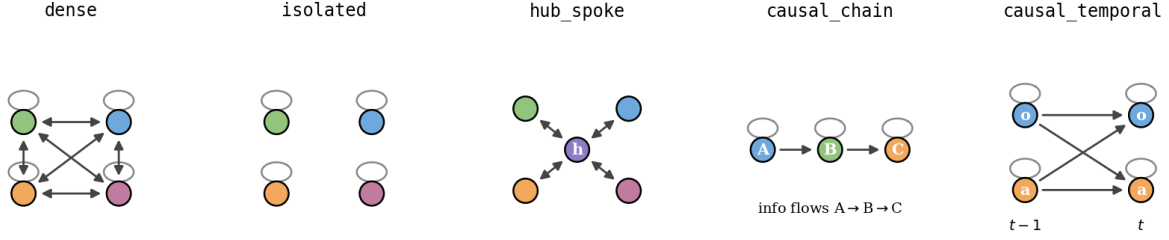


Figure 2: Topology constructors, drawn as information-flow graphs (arrows show where content moves; small loops are self-attention): *dense* (standard transformer), *isolated* (block-diagonal), *hub_spoke* (a shared coordinator), *causal_chain*, and *causal_temporal* (same-frame self + previous-frame cross, no future leakage). The interesting cases are structured DAGs that mirror a problem’s information flow.

to the most recent past anchor, *DROP* yields no edge, *INTERPOLATE* distributes weight $\propto 1/d^n$ over the $n+1$ nearest anchors. Execution is either per-edge cross-attention, $Y_{I_r} += w \text{softmax}(Q_{I_r} K_{I_s}^T / \sqrt{d}) V_{I_s}$ (with the edge’s *fn* substituting the kernel), or the monolithic equivalent, adding $\log M$ to dense attention logits with $\log 0 = -\infty$.

Loss participation is a weight vector. The declarations compile to $\omega_i = \sum_r \mathbf{1}[i \in I_r] \cdot \text{loss_weight}_r \cdot \mathbf{1}[\text{is_output}_r]$, and training minimizes the position-weighted denoising objective

$$\mathcal{L} = \mathbb{E}_{x_0, \epsilon, \sigma} \left[\frac{\sum_i \omega_i \|\hat{\epsilon}_\theta(x_\sigma)_i - \epsilon_i\|^2}{\sum_i \omega_i} \right],$$

where noise is applied only to output positions at $t \geq t_{\text{current}}$; context positions are clamped to their encodings, i.e. conditioning is inpainting-style [12, 13].

Reachability is the causal statement. Write the information-flow graph G with an arc $s \rightarrow r$ for every connection (r queries s , so content moves $s \rightarrow r$). One denoiser pass with L attention blocks moves information along paths of length $\leq L$ in G ; K sampling steps compose to KL . Region a can influence region b ’s denoised value iff G contains a directed path $a \rightarrow b$ —and if it does not, the independence is *exact*, by construction rather than by regularization: the canvas analogue of d -separation in a graphical model [15, 16].

2.4 A type system, literally

The claim that this is a type system is not decorative (Table 1, Fig. 3). `region_indices()` computes memory offsets; the topology is a calling convention; a serialized `CanvasSchema` (layout + topology + metadata, human-readable JSON) is a complete type signature. Two models that share a schema can exchange latent state directly—no tokenization, no re-encoding—because the schema fixes what every position *means*.

Across differing schemas, frozen semantic-type embeddings make modality compatibility computable (Fig. 6). Declaring a modality’s meaning as a string plus

Table 1: The type-system correspondence, item by item.

Type concept	Canvas equivalent	Implementation
Struct field (off-set+size)	Region bounds	<code>region_indices()</code>
Field annotation	period, loss_weight, sem. type	<code>RegionSpec</code>
Pointer / reference	Connection	<code>CanvasTopology</code>
Function signature	Topology + fn	<code>attention_ops()</code>
Type-directed codegen	Loss mask	<code>loss_weight_mask()</code>
ABI compatibility	Schema compat.	<code>compatible_regions()</code>
Coercion cost	Transfer distance	<code>transfer_distance()</code>

a fixed embedding from a declared model [17] turns compatibility from a judgment call into a distance: `transfer_distance(cam, depth)` ≈ 0.15 (bridgeable in 1–2 adapter layers) versus `transfer_distance(cam, joints)` ≈ 0.65 (a full adapter). This is contingent on the representation-stability hypothesis of §6.

2.5 Attention function types

Edges declare *how* information flows, not only *whether*. Seventeen registered function families span dot-product (`cross_`, `linear_`, `sigmoid_attention`), gating (`gated`), compression (`perceiver` [18], `pooling`), transfer (`copy`), state-space (`mamba` [19], `rwkv` [20]), convolution (`hyena` [21]), sparse (`local_`, `sparse_attention` [22]), and meta (`none`, `random_fixed`, `mixture`). Resolution order is `connection.fn` $\rightarrow$ `region.default_attn` $\rightarrow$ `global_cross_attention`. Each choice encodes a theory of the edge—pooling for a 12-D proprioception summary, `perceiver` to bottleneck 864 visual tokens into a thought region, `copy` for direct latent relay between agents. The schema declares intent; the executor picks implementation.

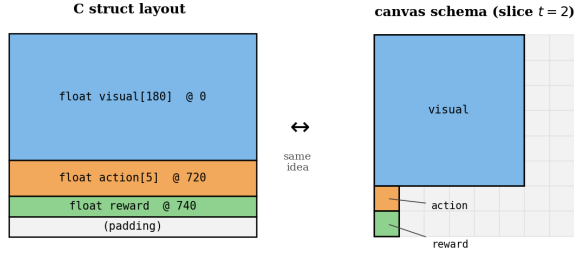


Figure 3: A C struct memory layout (left) and a canvas schema (right) are the same object: named fields at computed offsets, with a declared interpretation of every byte / position. The right panel is the $t=2$ slice of the layout in Fig. 1.

3 Compilation: from entity schema to canvas

Hand-placing rectangles does not scale, so the compiler accepts typed entity declarations—nested Python dataclasses; a pydantic surface is the same move—and flattens the entity/relationship graph they denote into layout + topology. Every nested type automatically receives a **coarse-grained field**: a compressed representation that bottlenecks all cross-level attention.

```
@dataclass
class Vehicle:
    __coarse__ = Field(4, 4) # 4x4 summary
    camera: Field = Field(8, 8)
    plan: Field = Field(2, 4)
    action: Field = Field(1, 4, loss_weight=2.0)

@dataclass
class Fleet:
    dispatch: Field = Field(4, 4)
    vehicles: list = dc_field(default_factory=list)

bound = compile_schema(
    Fleet(vehicles=[Vehicle() for _ in range(50)]),
    d_model=256)
```

Fifty vehicles under dense cross-attention would cost $O(50^2 \times \text{fields}^2)$ connections; under coarse-graining, each interacts through its 4×4 summary— $O(50 \times 16)$. Deep nesting chains the bottlenecks: in a world-model schema, the attention path from `us.macro.gdp` to `cn.macro.inflation` is forced through `us.macro` (coarse) $\rightarrow$ `us` (coarse) $\rightarrow$ `regime` $\rightarrow$ `cn` (coarse) $\rightarrow$ `cn.macro` (coarse). Each hop compresses, so hierarchical abstraction is not an emergent hope but a topological consequence. In effect the declared schema is a probabilistic graphical model over latent factors [15, 16], compiled into the attention structure of the denoiser rather than into a message-passing runtime.

The **program layer** (v2) adds process semantics per region: a *family* (observation / state / memory / residual / action), a *carrier* (deterministic / diffusive / filter / memory / residual), a *clock* (periodic or event-triggered, with composable firing rules such as `Or(periodic(4), on("err.prediction", gt=0.5))`), and a *compile mode* (runtime / freeze / constant / export). Connection operators are auto-derived from family pairs (observation $\rightarrow$ state = “observe”; state $\rightarrow$ action =

“act”); edges can carry triggers (skipped when a predicate over region statistics is false) and learned sigmoid gates. ProgramCompiler materializes deploy-time semantics: frozen regions get `requires_grad=False`, constants become buffers, and export regions write state dicts to disk. The stack is thus **entity types** $\rightarrow$ **canvas schema** $\rightarrow$ **wired dispatch** $\rightarrow$ **compiled deployment**, each layer inspectable, serializable, and diffable.

4 Worked designs

The abstraction earns its keep on problems whose causal structure is known but whose dynamics are not.

Robot manipulation. Vision-heavy, low-latency actions. `visual` uses full `cross_attention` for spatial reasoning; `proprio` (12-D joints) uses `linear_attention`; `visual` $\rightarrow$ `action` asks which patches matter, while `proprio` $\rightarrow$ `action` is a pooling summary. This is the configuration behind the empirical results of §5, trained on BridgeData V2 robot video [23].

Multi-agent coordination. A fleet (Fig. 5, left) declares per-vehicle self-attention with cross-edges only between neighbors and a shared dispatch hub; safety-critical air-traffic control (Fig. 5, right) declares 12 aircraft whose conflict geometry routes through a shared airspace field. The topology is the coordination protocol, written as data.

Hospital ICU: causal structure as a type hierarchy. The deepest schema in the library (Fig. 4): six patients with organ-level physiology, four nurses with fatigue and workload dynamics, a bureaucratic state (insurance, staffing, bed pressure), and family units. One `compile_schema` call turns that nested instance into 199 regions and 1,077 connections on a 26×26 canvas—the engineer writes the schema; the compiler does all placement. Each organ system is a nested type with per-field temporal periods that mirror clinical reality (`heart_rate` at `period=1`, `creatinine` at `period=24`, `delirium_risk` at `period=12`, `loss_weight=3.0`). A side effect worth stating on its own: the latent tensor becomes *pointable*. One can circle a run of blocks in Fig. 4 and say “that is `nurses[1]`”—interpretability not as post-hoc attribution but as a legend for memory, known before training starts. Crucially, the sepsis deterioration pathway—`renal.creatinine_rise` $\rightarrow$ `cardiovascular.MAP_drop` $\rightarrow$ `neurological.altered_consciousness` $\rightarrow$ `deterioration_risk`—is *declared*. A flat model learns the correlation; a canvas model is forced to route through the causal pathway, which is what makes the prediction interpretable and, by hypothesis, robust to distribution shift. Every field has a physiological reason to exist; every edge has a known biological pathway. The type system is not decoration—it is the domain knowledge, made executable.

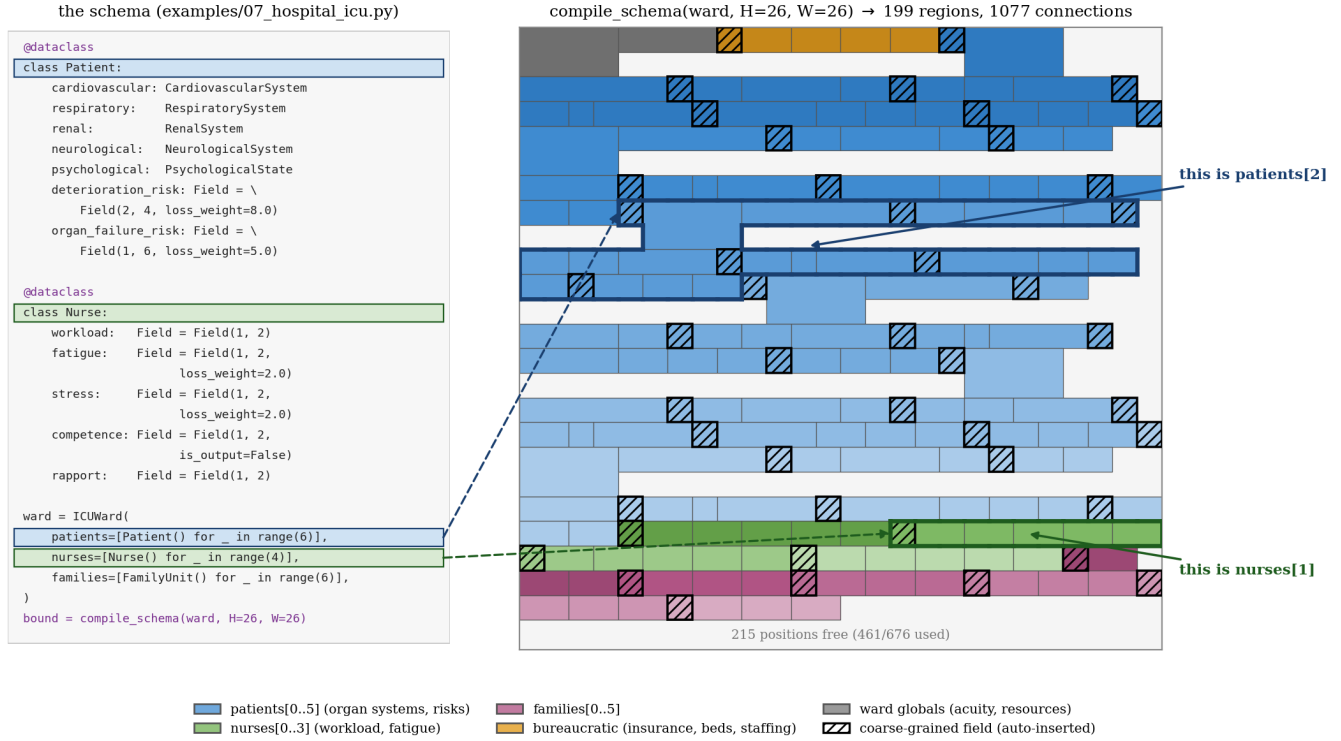


Figure 4: From schema to canvas, end to end. **Left:** the declaration—`Patient` and `Nurse` types (abridged) and the `ICUWard` instance passed to the compiler. **Right:** the compiler’s output—199 typed regions and 1,077 connections auto-packed onto a 26×26 canvas (68% utilization). Nobody placed a rectangle by hand. Hatched blocks are the coarse-grained fields the compiler inserts at every nesting level; all cross-entity attention is forced through them, so the hierarchy of the schema becomes a hierarchy of bottlenecks on the canvas. Dashed connectors trace two instances from source to allocation, and because the schema fixes what every position means, the latent tensor is *pointable*: the outlined runs of blocks are, exactly, `patients[2]` and `nurses[1]`.

Cortical world model. Declaration taken to its limit: 23 brain regions from the Destrieux atlas [24] wired by 42 known cortical pathways (ventral visual stream, the auditory-to-language route, executive control, default mode), across five families—observation (V1, A1, somatosensory), state (association areas), memory (temporal pole), action (motor cortex), and residual (prediction error). Trained to predict next-timestep cortical activation from Meta’s TRIBE brain-encoding predictions, the declared-topology model reaches $R^2 = 0.825$ on 135 features at 19.6% connectivity density, and a canvas EEG decoder hits 68.8% four-way accuracy (SVM 59.4%, chance 25%). The lesson from its ablations is sharp: on a low-dimensional version of the task a flat MLP wins; topology helps only once the state space is rich enough to exercise the declared pathways. **Topology is a convergence prior, not a capacity advantage.**

5 Empirical record

All results use a CogVideoX-2B backbone [6] on Bridge-Data V2 robot video [23]: 26 experiments, 236 training runs.

Looped attention. Orthogonal to the canvas, looped attention iterates frozen DiT blocks k times with zero-initialized learned iteration embeddings—so at initial-

ization the model is *exactly* the pretrained backbone, with no distribution shift—in the spirit of universal and recurrent transformers [25, 26]. From a 12-condition grid over loops $\times$ freeze level (Table 2):

Table 2: Action loss (lower better), 36 runs. Three loops wins at every freeze level; frozen 3-loop (350K params) beats every unfrozen condition (11.7M).

	Frozen (350K)	Half (3.7M)	Unfrozen (11.7M)
1 loop	0.121	0.115	0.108
3 loops	0.073	0.107	0.088
4 loops	0.104	0.137	0.124

Recurrence beats depth: $1.73\times$ **parameter efficiency** ($p < 0.001$). Freeze level does not affect action loss at all (marginals 0.109 vs. 0.108, $p = 0.72$); it only affects video-generation quality (an 8–9 $\times$ gap on diffusion loss). Loop representations converge toward fixed points (cosine similarity to loop 1: $0.926 \rightarrow 0.996$; token velocities decay exponentially, action tokens fastest).

Reading the data at the right altitude. These are 2B-parameter, single-domain, small-data experiments, and we deliberately avoid narratives the data cannot yet

carry. Three findings calibrate what the architecture does at this scale, and each one *informs the design* rather than undermining it:

1. *Where looping’s benefit comes from.* At this scale the measurable benefit of looped attention is weight-sharing regularization—fixed-point convergence, lower variance, the 1.73× efficiency above—rather than detectable iterative reasoning (three independent nulls). Whether reasoning-like refinement emerges with larger canvases, longer horizons, and more compute is exactly the question the architecture is built to test; we decline to claim it early.
2. *Structure must be declared, not hoped for.* Merely co-locating modalities on a flat, densely-attending canvas did not improve joint prediction—in our runs it degraded it by 19% ($p < 0.0001$). We read this as evidence that binding, like the rest of the structure, has to be declared rather than left to emerge: the capability came from the typed multi-encoder/decoder interface, and co-location alone did not add to it. The natural follow-up—the same canvas with and without declared connectivity—is the next experiment.
3. *Canvas design is forgiving.* Loss is nearly insensitive to token allocation ($\alpha = 0.011$: doubling a region’s tokens moves loss 0.8%). Geometry can therefore serve legibility and engineering convenience without taxing accuracy—good news for a discipline whose premise is that humans draw the layout.

6 Open problems

Representation stability (the linchpin). Everything interoperable—schema-mediated latent exchange, transfer-distance calibration, plug-and-play modalities, swap-the-policy-keep-the-perception composition—rests on a platonic-representation-style claim [27]: that identical declared structure induces predictably aligned latent geometry across seeds, datasets, and backbones. Plausible, unproven. The test battery is specified: seed-stability CKA across corresponding regions [28]; topology→specialization ablations on fixed data; cross-model grafting cost; and regression of measured transfer cost against semantic-embedding distance.

Binding. The flat-canvas co-residence penalty (§5) says implicit binding is not free. Whether *declared topology*—the same canvas with authored connectivity—recovers or exceeds independent-model performance is the decisive ablation, and one on which the framework makes a concrete, testable prediction.

Scale. All evidence is at 2B parameters and robot-video scale. Whether declared macrostructure keeps paying as data grows, or is eventually washed out [7, 29], is open. The bet is asymmetric: schemas cost

nothing to author, compile onto pretrained backbones, and can be relaxed edge-by-edge (`fn="none" → dense`) if they fail to bind.

Learned topology. Today the graph is authored. The natural continuation—propose-edges / prune-edges under sparsity pressure, with the declared schema as prior rather than hard constraint—would close the loop back to Chollet’s synthesis route, with the canvas as the substrate both routes share.

7 Related work

Diffusion policy [14] establishes observation-conditioned action diffusion—the two-node canvas. Unified multimodal DiTs [5, 6] share one latent sequence but leave structure implicit, to be discovered from surface gradients: the exact move canvas engineering replaces with declaration. Structured-attention mechanisms—sparse patterns [22], Perceiver bottlenecks [18], state-space and long-convolution mixers [19–21]—supply the executors; the contribution here is not a mechanism but an authorable, compilable, serializable *schema layer* over them. Probabilistic graphical and structural-causal models [15, 16] are the ancestors: a canvas topology is a causal graph whose message passing is implemented by a pretrained denoiser’s attention rather than by belief propagation. World models [8–10] motivate the target application. Looped and universal transformers [25, 26] study recurrence as adaptive compute; at the scales we have tested, its practical value on frozen video backbones is weight-sharing regularization.

8 Conclusion

The neurosymbolic program stalls in production because it usually demands a new runtime at the neural/symbolic boundary. Canvas engineering dissolves the boundary instead: the symbolic artifact is a schema—a type system for latent computation—and its execution is masked attention inside a stock diffusion transformer. You declare the causal macrostructure of thought; SGD fills in the rest. The empirical record is early and we present it at face value: 1.73× parameter efficiency, a 350K-parameter configuration that beats 11.7M, and calibration results that tell us where structure must be declared rather than hoped for. The decisive experiments—declared topology versus flat co-residence, representation stability across seeds and backbones, behavior at scale—are specified and awaiting compute, and we would rather run them than speculate ahead of them. What stands regardless is the abstraction. Once latent structure is *declarable*, it is versionable, diffable, compilable, and shareable—and structure you can ship is the only structure that will ever out-compete paying for more data.

Artifacts. [pip install canvas-engineering](#) (Apache 2.0), published under [CommandAGI](#). Code:

github.com/commandAGI/canvas-engineering; documentation, runnable examples, and design notes: commandagi.github.io/canvas-engineering. Full experiment archive and the companion empirical paper (“Looped Attention in Video Diffusion Transformers: 26 Experiments on What Works, What Doesn’t, and Why,” Valdez & Claude Opus 4.6) at github.com/JacobFV/recursive-omnimodal-video-action-model.

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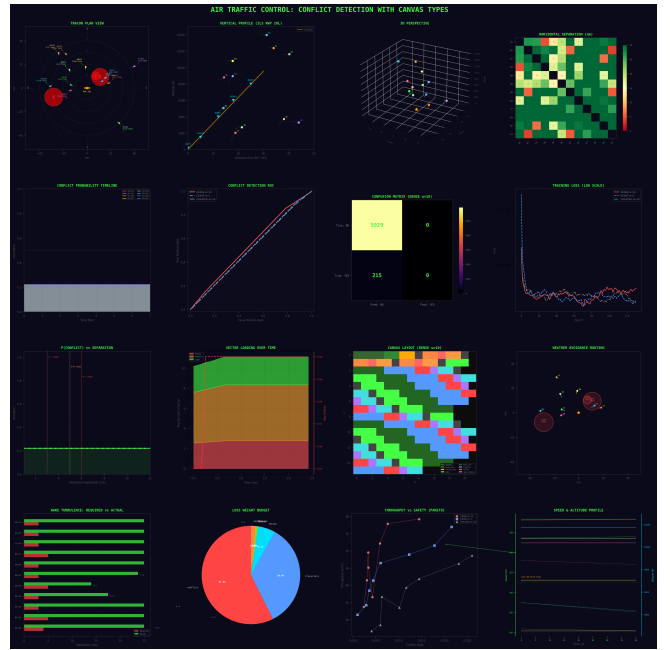
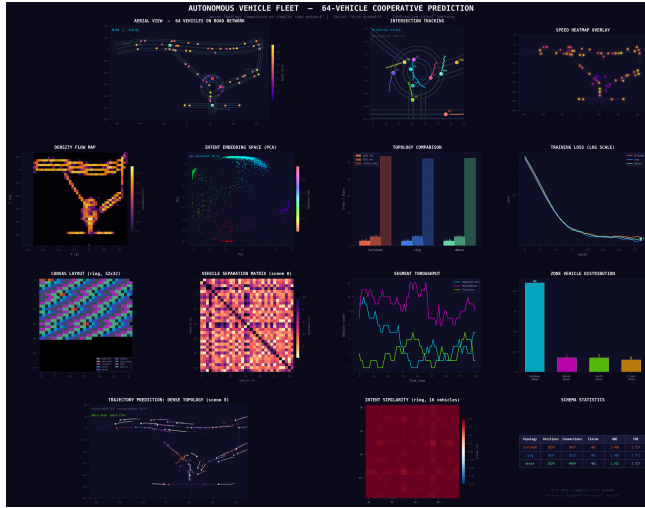


Figure 5: Two multi-agent canvases. **Left:** cooperative vehicle-fleet trajectory prediction, per-vehicle regions coordinating through a shared dispatch hub via coarse-grained summaries. **Right:** safety-critical air-traffic control with 12 aircraft and declared conflict-detection routing. In both, the attention topology is the coordination protocol expressed as data, not a pattern the network must rediscover.

Semantic Embedding Space — Transfer Distance

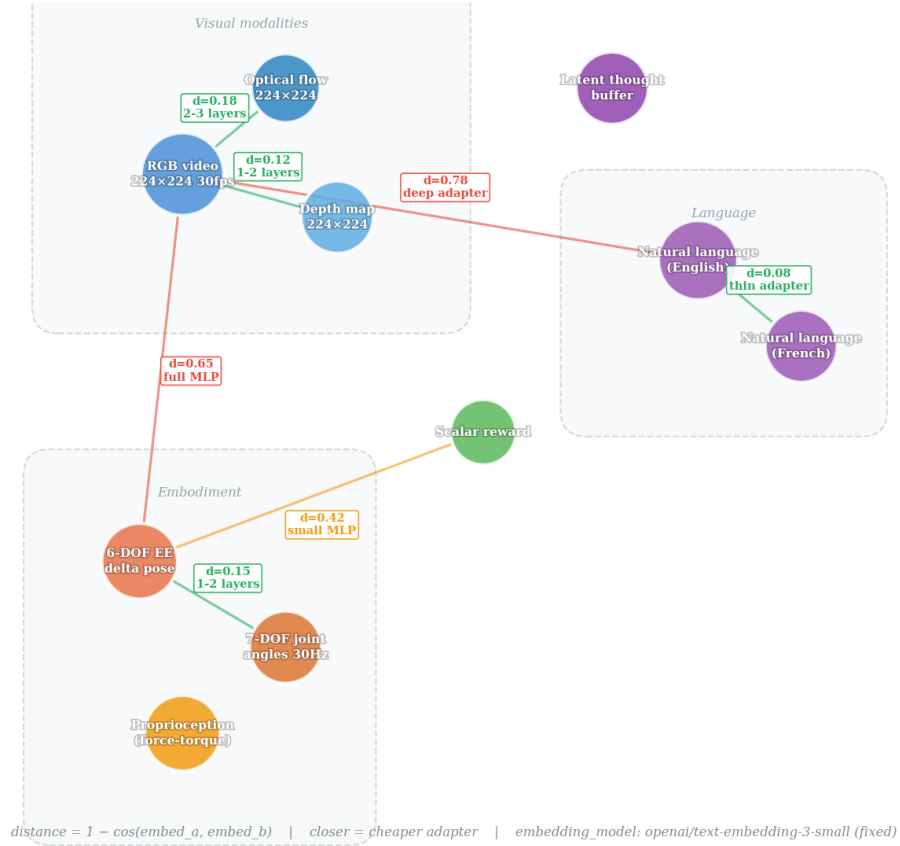


Figure 6: Semantic-type embedding space. Each region declares its modality as a string plus a frozen embedding from a fixed model; cosine distance between embeddings estimates the real cost of bridging two modalities (adapter layers, data). Camera↔depth is cheap; camera↔joint-angles is expensive. This makes modality compatibility a computable quantity rather than a judgment call—conditional on representation stability (§6).